

tracking app — Identity & History (the append-only memory)

Dual-register guide. Canonical: [docs/production/TRACKING.md](#) · [ADR-0004](#) · barcode permanence: [ADR-0010 §3](#).

Purpose & business responsibility

Do kaam, dono “yaad rakhna”:

1. **History** — har important cheez ka timeline (roll, Adda, product, settlement events) — append-only, kabhi edit nahi.
2. **Barcode identity** — har piece ka number ((adda, seq) ranges).

R1 boundary (locked): **tracking owns identity + scan history; production owns quantity truth**. Tracking imports NO production code (string FKs only).

Models

Model	Business	DB meaning / example
AddaHistory	one event in an Adda's life (stage moved, reopened, SETTLEMENT_FINALIZED/REVERSED/SUPERSEDED...)	append-only; metadata JSONB for event details. Row: adda=3-PATTI-001, change_type=settlement_finalized, actor=manager1, metadata={ 'reference': 'ADST-0003' }
ClothRollHistory	roll movements/edits (incl. cost_per_kg corrections)	append-only
ProductHistory	product master changes	append-only
BarcodeBatch	ONE contiguous seq range per (adda,color,size)	3-PATTI-001 Red Size-1 → seq 1..60 — piece identity = (adda, seq), derivable without a Piece table; printed payloads are PERMANENT (ADR-0010 §3)
BatchBarcode / BarcodeExportBatch	rendered/export bookkeeping	see docs/tracking/EXPORTS.md

How data flows through this app

```
any meaningful change in production/raw_materials/expense
| (the writer service calls...)
▼
history_service.log_adda / log_roll / log_product ← SOLE writer (gate: rule 5)
▼
*History rows (INSERT only) → timeline accordions + future Adda-360 (G4)

barcode_generation stage (archetype E)
▼
barcode_service → BarcodeBatch ranges → QR print sheets
```

What breaks if bypassed: history written ad-hoc loses the “who/when/why” discipline (and signals are banned — ADR-0001, so there is no magic catcher); a second barcode numbering scheme would orphan every physically printed garment — the one thing no migration can fix.

Views: barcode dashboard, export flows, print sheet (standalone A4 HTML).

Common mistakes

1. Never write `*History` outside `history_service`.
2. Never re-number barcodes; future Piece/ScanEvent models must be born FROM existing ranges (TM-2 / Barcode review).
3. Don't put production logic here — string-FK boundary is deliberate.

Django Learning Notes

- **String FKs** (`'production.Adda'`): dependency direction control — the app below imports nothing from above (import-linter contract material).
- **JSONField metadata**: display-grade detail without schema churn; promote to columns only if filtering is ever needed.
- **Append-only pattern**: no `updated_at` semantics on history — a fix is a NEW row that says what changed.

Real factory example

3-PATTI-001 ki timeline kholo: started → layering complete → cutting complete (by utest) → barcode batch 1...105 → SETTLEMENT_FINALIZED ADST-0001 → SETTLEMENT_SUPERSEDED → SETTLEMENT_FINALIZED ADST-0003. Har garment pe jo QR laga hai (`3-PATTI-001 · 37`) wo hamesha ke liye usi piece ka naam hai.